Ahmad Samim Hamza

Senior Software & AI Engineer

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ABOUT ME

AI & Senior Software Engineer and Master's student in Artificial Intelligence at the University of Jyväskylä, with 5 years of experience building intelligent applications and the reliable software systems around them. At Konecranes, I owned the end-to-end delivery of an enterprise RAG and multi-agent assistant using AWS Bedrock (Agents, AgentCore), LangGraph, and LangChain—evaluating architectural trade-offs, integrating PostgreSQL feedback pipelines, and deploying on AWS (Amplify, ECS, RDS) with CloudWatch monitoring. Earlier in my career, I architected and delivered production ERP, logistics, and e-commerce platforms and led a team of eight engineers. I combine strong software engineering foundations (Python, FastAPI, TypeScript, NestJS, Next.js, PostgreSQL, Docker, CI/CD) with daily AI-assisted engineering workflows using Claude Code and Cursor to ship tested, production-grade code rapidly.

SKILLS

AI, ML & Agentic Systems: LangGraph | LangChain | AWS Bedrock (Agents & AgentCore) | Multi-Agent Orchestration | MCP | Corrective RAG (CRAG) | Hybrid Search (BM25 + Dense Vectors) | RRF & Cross-Encoder Reranking | Vector Stores (Qdrant, ChromaDB, S3) | LLMOps & Evaluation

AI-Assisted Engineering: Claude Code | Cursor | Antigravity | Prompt & Spec Engineering

Backend & Data Engineering: Python (FastAPI, AsyncIO, Django) | TypeScript/JavaScript (NestJS, Node.js, Express) | PostgreSQL (Schema Design, Query Optimization, pgvector) | Semantic Web (Ontologies, RDF, OWL, SPARQL) | REST APIs | PHP (Laravel) | MySQL | SQLite | Redis

Frontend: Next.js | React | TypeScript | Tailwind CSS | Material UI | WebSockets | Vue.js | Nuxt.js

Cloud, Platforms & DevOps: AWS (Bedrock, ECS, Amplify, RDS, S3, Lambda, API Gateway, CloudWatch, ECR) | Azure (AI Foundry, AI Search) | Docker | Kubernetes | CI/CD (GitHub Actions) | Linux

WORK EXPERIENCE

Software Development Specialist (AI/ML) – Fixed-Term Summer Contract 05/2026 – 08/2026
Konecranes Hyvinkää, Finland

- Owned end-to-end design, development, and AWS deployment of a production enterprise AI assistant, taking the solution from initial problem discovery and prototyping to full documentation and handover to the internal team.
- Built and benchmarked multiple production-ready RAG variants on AWS Bedrock (Agents & AgentCore), LangChain, and LangGraph over 100+ internal XML, PDF, and XLSX documents indexed in an S3 vector store, evaluating trade-offs in retrieval accuracy, response latency, and maintainability.
- Architected a multi-agent system on AWS Bedrock with a supervisor agent orchestrating task-specific collaborator agents across enterprise knowledge sources.
- Integrated the RAG and agentic backends into a unified Next.js and FastAPI application behind a shared conversational UI, using PostgreSQL (AWS RDS) for user accounts, session management, and a human-in-the-loop feedback pipeline to evaluate three RAG variants across internal teams.
- Developed the entire Next.js and FastAPI application using Claude Code and Cursor as primary AI-assisted coding environments, accelerating iteration speed while enforcing strict production code quality.
- Deployed the platform on AWS (Amplify, ECS, RDS), collaborated with stakeholders to align agentic workflows with business operations, and monitored post-deployment health via CloudWatch logs and application error tracking to identify and resolve issues affecting retrieval accuracy and response latency.

Senior Software Engineer 05/2024 – 07/2025
Ramaki Group of Companies Kabul, Afghanistan

- Led and coordinated a team of eight engineers in the end-to-end development of Dara.af , a production e-commerce platform supporting hundreds of product listings and daily customer orders using Laravel, PostgreSQL, React, and Next.js.
- Designed and optimized PostgreSQL database schemas for complex transactional and inventory data.
- Built RESTful APIs and wrote unit, integration, and feature tests to ensure system reliability and data integrity.
- Conducted code reviews, established engineering standards, and mentored junior developers.

Software Engineer 09/2022 – 04/2024
Peace Global Logistics Kabul, Afghanistan
- Collaborated with logistics, finance, and operations domain experts to translate complex supply-chain and shipment tracking workflows into production ERP modules using NestJS, React, Next.js, and PostgreSQL.
- Led the architectural migration and database schema redesign of core ERP features (finance, logistics, inventory, and audit logging) into a modern NestJS, Next.js, and PostgreSQL stack.
- Optimized complex PostgreSQL queries for high-volume order and shipment data and built reusable React + Tailwind dashboard components.

Instructor (Python & Machine Learning) 04/2022 – 10/2022
Kabul University Kabul, Afghanistan
- Taught Python programming and machine learning fundamentals to freshman and sophomore undergraduate students.

Full Stack Developer 07/2021 – 09/2022
Smart Friqi Kabul, Afghanistan
- Built production ERP modules using Laravel, Nuxt.js, and PostgreSQL; implemented real-time operational features using WebSockets; and developed interactive dashboards for business management and operations monitoring.

PROJECTS

Enterprise RAG & Multi-Agent Assistant (Konecranes) 05/2026 – 08/2026
Tech: AWS Bedrock (Agents, AgentCore), LangChain, LangGraph, FastAPI, Next.js, PostgreSQL, AWS Amplify/ECS/RDS, CloudWatch, Claude Code
- Internal enterprise knowledge assistant over 100+ heterogeneous documents (XML, PDF, XLSX), benchmarking multiple RAG variants behind a shared conversational UI with multi-agent supervisor orchestration, a PostgreSQL evaluation pipeline, and CloudWatch monitoring.

CRAG Hybrid Retriever — Open-Source Python IR & Verification Engine 10/2026 – Present
Tech Stack: Python, BM25Okapi, Dense Vector Search, Reciprocal Rank Fusion (RRF), FlashRank (ONNX), Pydantic v2, uv, Pytest, GitHub Actions CI
- Engineered an open-source hybrid search library combining BM25 lexical search and dense vectors via Reciprocal Rank Fusion (RRF, k=60) to eliminate semantic drift.
- Integrated ONNX cross-encoder reranking (FlashRank) and a self-reflective CRAG engine that grades evidence confidence and dynamically reformulates ambiguous queries.
- Built production-grade architecture with strict Pydantic v2 typing, fast uv packaging, 100% pytest test coverage, and automated GitHub Actions CI.

Dara.af E-Commerce Platform (dara.af) 05/2024 – 07/2025
Tech: Laravel, PostgreSQL, React, Next.js, REST API
- Led an 8-engineer team through PostgreSQL schema design, REST API development, and full-stack delivery for a production e-commerce platform supporting daily customer orders.

PGL ERP System (pglsystem.com) 09/2022 – 04/2024
Tech: NestJS, React, Next.js, PostgreSQL, REST API
- Built logistics, shipments, HR, finance, and inventory modules for Peace Global Logistics and designed APIs optimizing data flow across the enterprise platform.

EDUCATION

Master of Science in Artificial Intelligence 08/2025 – Present
University of Jyväskylä Jyväskylä, Finland
- Master's Thesis (In Progress): Exploring LLM-assisted validation for multi-objective optimization in Finnish forest planning.

Bachelor of Computer Science (Software Engineering) 2017 – 2021
Kabul University Kabul, Afghanistan

LANGUAGE SKILLS

- English (Fluent)
- Persian (Native)
- Finnish (A1 Level – Actively Learning)